

4. ReactJS-HOL

1. Explain the Need and Benefits of Component Life Cycle

What is a Component Life Cycle?

The **Component Life Cycle** is the sequence of stages a React component goes through from the time it is created until it is removed from the DOM. React provides **lifecycle methods (hooks)** that allow developers to execute code at different stages of a component's life.

Need for Component Life Cycle

- To perform actions when a component is created.
- To fetch data from APIs.
- To update the UI when state or props change.
- To optimize application performance.
- To clean up resources before a component is removed.

Benefits of Component Life Cycle

- Enables data fetching from servers.
- Helps manage component state efficiently.
- Improves application performance.
- Makes debugging easier.
- Allows cleanup of timers, subscriptions, and event listeners.
- Provides better control over component behavior.

2. Identify Various Life Cycle Hook Methods

React class components have three major phases of the lifecycle.

A. Mounting Phase (Component is Created)

- **constructor()**
 - Initializes state and binds methods.
- **static getDerivedStateFromProps()**
 - Updates state based on props.
- **render()**

- Returns the JSX to display.
 - **componentDidMount()**
 - Called after the component is added to the DOM.
 - Commonly used for API calls.
-

B. Updating Phase (Component is Updated)

- **static getDerivedStateFromProps()**
 - **shouldComponentUpdate()**
 - Decides whether the component should re-render.
 - **render()**
 - **getSnapshotBeforeUpdate()**
 - Captures information before DOM updates.
 - **componentDidUpdate()**
 - Executes after the component updates.
-

C. Unmounting Phase (Component is Removed)

- **componentWillUnmount()**
 - Performs cleanup before the component is removed.
 - Used to clear timers, cancel API requests, and remove event listeners.
-

D. Error Handling Phase

- **componentDidCatch()**
 - Catches JavaScript errors in child components.
 - Displays fallback UI or logs errors.
-

3. List the Sequence of Steps in Rendering a Component

The rendering process follows the component lifecycle.

Mounting Sequence

Constructor()



getDerivedStateFromProps()



render()



componentDidMount()

Updating Sequence

getDerivedStateFromProps()



shouldComponentUpdate()



render()



getSnapshotBeforeUpdate()



componentDidUpdate()

Unmounting Sequence

componentWillUnmount()

Error Handling Sequence

Error Occurs



componentDidCatch()

Summary Table

Lifecycle Phase	Lifecycle Methods
Mounting	constructor(), getDerivedStateFromProps(), render(), componentDidMount()
Updating	getDerivedStateFromProps(), shouldComponentUpdate(), render(), getSnapshotBeforeUpdate(), componentDidUpdate()
Unmounting	componentWillUnmount()
Error Handling	componentDidCatch()

These answers are appropriate for your ReactJS Hands-on Lab record and cover the objectives listed in the document.